

MODULE 03 — ASSIGNMENT

Software Testing Life Cycle (STLC) with AI

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Course: AI Driven SQA — Manual & Automation Testing | OSTAD

Q1. Test Plan — Online Shopping Website

Objective: confirm that the shopping website's core customer journey — registration, login, product search, cart, and checkout — is functionally correct, secure against basic misuse, and usable across common browsers and devices.

Test Objectives

- Confirm business-critical shopping functions against approved requirements.
- Validate positive, negative, boundary and error-handling scenarios.
- Confirm critical customer journeys remain stable across supported browsers and devices.

1. Scope & Coverage

Test Area	Coverage
Login	Valid/invalid credentials, required-field checks, session handling, logout, lockout behaviour.
Registration	Field validation, password rules, duplicate-account handling, terms acceptance, verification flow.
Product Search & Browse	Keyword/category search, filters, sorting, empty-result handling.
Cart	Add/remove/update items, quantity limits, price recalculation, stock checks.
Checkout & Payment	Address entry, delivery options, discount codes, payment method selection, order confirmation.
Order Confirmation	Order ID generation, summary accuracy, confirmation messaging.

Out of scope: third-party payment provider internals, production server administration, and features not yet released to QA.

2. Test Strategy

Technique	Purpose	Priority
Functional Testing	Confirm each feature behaves as specified for valid, everyday customer actions.	High
Negative & Boundary Testing	Use invalid, missing, extreme, or edge-case inputs to confirm the system fails safely and clearly.	High
Integration Testing	Verify handoffs between login, catalog, cart, checkout, and payment components work together correctly.	High
End-to-End Testing	Walk a full customer journey — browse to registration to checkout — to confirm the whole flow holds together.	High
Regression Testing	Re-run priority tests after each fix or release to confirm nothing that worked before has broken.	High
Usability Testing	Check that labels, error messages, and navigation are clear enough for a first-time shopper.	Medium
Compatibility Testing	Confirm the site behaves consistently across supported browsers and screen sizes.	Medium
Security Spot-Checks	Probe login/registration inputs for basic injection and data-exposure risks.	Medium
AI-Assisted Test Design	Use AI to draft scenario lists and edge cases quickly; a QA engineer reviews and adjusts every item before use.	Medium

3. Test Environment & Data

Item	Planned Setup
Test Environment	Isolated QA/staging environment, separate from production data and traffic.
Browsers	Latest stable Chrome, Firefox, and Edge, plus one mobile browser (Chrome Android or Safari iOS).
Devices	One desktop resolution and one representative mobile viewport, per the supported device matrix.
Test Accounts	Valid registered user, newly created user, duplicate-email user, and a locked/invalid-credential account.
Test Data	Valid and invalid addresses, in-stock and out-of-stock products, valid and expired discount codes, sandbox payment cards.
External Services	Sandbox/mock payment gateway; no real transactions are processed during testing.

4. Entry & Exit Criteria

Gate	Criteria
Entry Criteria	Build is deployed to QA; requirements are stable enough to test; required test accounts and data exist; test environment is reachable.
Exit Criteria	All high-priority test cases executed; no open blocker or critical defects; failed cases triaged and either fixed or accepted as known issues; results documented for sign-off.

5. Roles & Responsibilities

Role	Responsibility
QA Engineer	Writes and executes test cases/checklists, logs defects, retests fixes, reports status.
QA Lead	Owns test strategy and coverage decisions, manages risk trade-offs, reports to stakeholders.
Developer	Fixes reported defects, clarifies implementation behaviour, supports root-cause investigation.
Product Owner	Clarifies requirements, prioritises business risk, signs off on acceptance criteria.
DevOps / Environment Owner	Keeps the QA environment, test data, and deployment pipeline available and stable.

6. Risks & Mitigation

Risk	Impact	Mitigation
Payment gateway sandbox is unavailable	Blocks end-to-end checkout validation.	Use a mock/stub response for the payment step; test the surrounding flow independently.
Test environment is unstable	Causes false failures and wastes execution time.	Log environment incidents separately from product defects; re-run only after a confirmed fix.
Requirements are incomplete or ambiguous	Leads to disagreement on expected results.	Confirm expected behaviour with the Product Owner before finalising test cases.
Limited time for regression	Existing features may silently break.	Prioritise regression around the highest-traffic flows: login, cart, and checkout.
Test data becomes stale or reused incorrectly	Produces misleading pass/fail results.	Maintain a small, controlled, reset-able set of test accounts and products.

7. Defect Management

- Every defect is logged with steps to reproduce, expected vs. actual result, severity, and supporting evidence (screenshot/log).
- Defects are triaged jointly with development and the Product Owner before a fix is scheduled.
- Every fix is retested, and a short regression pass covers the surrounding functionality.
- Release readiness is judged by unresolved risk, not just by raw defect count.

AI use & human review: AI tools may be used to draft test ideas, edge cases, or documentation faster. However, interpreting requirements, judging real business risk, validating the test environment, and signing off on quality remain the QA engineer's responsibility — not the AI's.

Q2. Test Checklist

Each feature below is covered by 20 validation points using the required columns: Test Area, Validation Point, and Expected Result.

A. Login Checklist

ID	Test Area	Validation Point	Expected Result
LG-01	Required fields	Submit the login form with both fields empty.	Both fields show a required-field message; no login attempt is sent.
LG-02	Valid login	Enter a correct, registered email/username and password.	User is authenticated and lands on the expected post-login page.
LG-03	Wrong password	Enter a valid email with an incorrect password.	Login is rejected with a generic error, without confirming which field was wrong.
LG-04	Unregistered account	Enter an email that has never been registered.	Login fails with the same generic message used for a wrong password.
LG-05	Invalid email format	Enter text that is not a valid email pattern.	Format error appears before any server round-trip, where client-side validation exists.
LG-06	Empty password only	Leave the password blank with a valid email entered.	Password-required message appears; login is blocked.
LG-07	Password masking	Type a password into the field.	Characters are masked by default (dots/asterisks).
LG-08	Show/hide password toggle	Use the eye icon if the field provides one.	Toggling reveals/hides the password without altering the typed value.
LG-09	Case sensitivity	Enter the registered email in a different letter case.	Email is treated as case-insensitive (standard expectation) unless documented otherwise.
LG-10	Leading/trailing spaces	Enter the email with extra spaces before or after it.	Input is trimmed automatically, or a clear validation message appears.
LG-11	Repeated failed attempts	Submit wrong credentials several times in a row.	Account lockout or throttling behaviour triggers per the security policy, if one exists.
LG-12	SQL/script injection attempt	Enter a string like ' OR 1=1-- into the login fields.	Input is treated as plain text; no error page or unexpected behaviour occurs.
LG-13	Session after logout	Log out, then press the browser Back button.	The previous authenticated page is not shown; login is required again.
LG-14	Direct URL access	While logged out, paste a direct URL to an account-only page.	User is redirected to the login page instead of seeing protected content.
LG-15	Remember-me option	Select "Remember Me" (if available), close and reopen the browser.	Session persists or expires exactly as the feature is documented to behave.
LG-16	Session timeout	Stay idle past the configured timeout, if one is defined.	Session expires and the next action requires re-authentication.
LG-17	Keyboard-only navigation	Tab through the login form without using a mouse.	Focus order is logical; the form can be fully completed and submitted via keyboard.
LG-18	Field labels for screen readers	Inspect email/password fields with a screen reader or accessibility checker.	Each field has an associated, descriptive label.
LG-19	Mobile layout	Open the login page on a small mobile viewport.	Fields, buttons, and error text remain fully visible with no overlap.
LG-20	Auto fill behavior	Use the browser's saved-password auto fill.	Auto filled credentials populate correctly and can still be edited before submit.

B. Registration Checklist

ID	Test Area	Validation Point	Expected Result
RG-01	Required fields	Submit the form with all mandatory fields empty.	Each mandatory field shows its own required-field message.
RG-02	Valid registration	Fill all fields correctly with a new email and submit.	Account is created, or a verification step begins, exactly as designed.
RG-03	Invalid email format	Enter an email missing the @ symbol or domain.	Format validation blocks submission before an account is created.
RG-04	Duplicate email	Register using an email that already has an account.	Registration is blocked with a clear "already registered" message.
RG-05	Weak password	Enter a password that fails the stated complexity rule.	The specific unmet rule is shown; submission is blocked.
RG-06	Password/confirm mismatch	Enter different values in Password and Confirm Password.	Mismatch is flagged; the account is not created.
RG-07	Password/confirm match	Enter identical, valid values in both password fields.	Validation passes and registration proceeds.
RG-08	Minimum length boundary	Enter a password exactly at, and one character below, the minimum length.	The below-minimum case is rejected; the exact-minimum case is accepted.
RG-09	Maximum length boundary	Enter a name/email far longer than the field is expected to allow.	Input is truncated, rejected, or safely handled — never causes a server error.
RG-10	Invalid characters in name	Enter numbers or symbols into a name field, if only letters are expected.	Field either blocks the invalid characters or the rule is confirmed as not enforced.
RG-11	Phone number format	Enter a phone number with letters or an incorrect number of digits.	Validation catches the malformed number before submission.
RG-12	Terms not accepted	Leave the Terms & Conditions checkbox unticked and submit.	Registration is blocked until the box is checked.
RG-13	Terms accepted	Tick the checkbox with otherwise valid data and submit.	Registration proceeds without the terms error.
RG-14	Email/OTP verification	Complete registration where a verification step is required.	Account is fully active only after the verification code/link is confirmed.
RG-15	Expired or wrong verification code	Enter an expired or incorrect verification code.	Verification fails with a message and a clear way to request a new code.
RG-16	Injection-style input	Enter script-like text into a text field.	Text is stored/displayed literally; no script executes.
RG-17	Copy-paste with extra spaces	Paste a name or email with leading/trailing whitespace.	Whitespace is trimmed automatically or flagged clearly.
RG-18	Password field exposure	Watch the password field while typing and on page inspection.	Value is masked on screen and not visible in plain text in the page source.
RG-19	Mobile layout	Complete registration on a small mobile viewport.	All fields, validation messages, and buttons remain usable without clipping.
RG-20	Keyboard accessibility	Complete the entire form using only Tab and Enter.	Every field and the submit button are reachable in a logical order.

C. Checkout Checklist

ID	Test Area	Validation Point	Expected Result
CO-01	Checkout with valid cart	Open checkout with an in-stock item already in the cart.	Checkout loads with the correct item, quantity, and price.
CO-02	Empty cart checkout	Try to reach checkout with an empty cart.	Checkout is blocked, or the customer is redirected with a clear empty-cart message.
CO-03	Missing address	Submit the order without entering a shipping address.	Required-field validation appears; the order is not placed.
CO-04	Valid address	Enter a complete, correctly formatted shipping address.	Address is accepted and displayed correctly in the order summary.
CO-05	Incomplete address	Enter an address missing a required part (e.g. postal code).	Validation identifies the missing field before submission.
CO-06	Delivery option selection	Choose an available delivery method.	Selected option and its fee/timeframe are reflected in the order total.
CO-07	Delivery fee recalculation	Switch between two different delivery options.	Order total updates immediately to match the newly selected option.
CO-08	Order summary accuracy	Compare the checkout summary against the actual cart contents.	Items, quantities, and unit prices match exactly.
CO-09	Quantity change at checkout	Increase or decrease an item's quantity, if editable at this step.	Subtotal and stock availability are re-validated correctly.
CO-10	Item goes out of stock mid-checkout	Simulate an item becoming unavailable before payment.	Customer is clearly notified; the unavailable item cannot be purchased.
CO-11	Valid discount code	Apply a currently valid promotional code.	Discount is applied correctly and the new total is shown immediately.
CO-12	Invalid or expired discount code	Apply a code that is wrong, expired, or already used.	Code is rejected with a clear message; the total remains unchanged.
CO-13	Payment method selection	Select each available payment method in turn.	The correct next steps/fields appear for whichever method is selected.
CO-14	Payment failure handling	Use a test card or scenario designed to fail.	Order is not confirmed; customer sees a clear failure message and can retry.
CO-15	Successful payment	Complete payment using an approved sandbox success scenario.	Payment succeeds and the flow proceeds to order confirmation.
CO-16	Double-click on Place Order	Click or tap the Place Order button rapidly more than once.	Only one order is created; the second click is safely ignored.
CO-17	Session expires during checkout	Let the session time out, then attempt to place the order.	Customer is prompted to re-authenticate; no order is created without a valid session.
CO-18	Grand total accuracy	Manually add item total, delivery fee, discount, and tax (if any).	Displayed grand total matches the manual calculation exactly.
CO-19	Order confirmation details	Complete a successful order.	Confirmation page/email shows a unique order ID and an accurate summary.
CO-20	Mobile checkout usability	Complete the full checkout flow on a mobile viewport.	Every step remains legible and operable without horizontal scrolling or overlap.

Final QA note: this checklist goes beyond the happy path — it includes negative, boundary, security-lite, and accessibility checks. Before execution, map each point to the site's actual approved requirements and supported browser/device list.